

## **Assignment 10: Explainable Artificial Intelligence Toward Responsible AI**

### **1. Why advocate for transparency or explainability in an ML or AI system?**

I would advocate for transparency and explainability because they help people understand why a system behaves the way it does, which increases trust and confidence. The assigned article explains that explainability serves different audiences in different ways, including helping developers debug, helping domain experts evaluate decisions, supporting end users and enabling regulators to check compliance (Barredo Arrieta et al., 2020). Figure 2 illustrates that explainability consistently supports two goals: enabling model understanding and meeting regulatory expectations. This reinforces that explainability is not optional but a foundational part of making information systems work for the people who rely on them.

As a student in the UW iSchool, I see explainability as an essential part of responsible information practice. Information systems shape people's opportunities, access and outcomes. When a model influences something that matters in someone's life, that person should not feel powerless or confused about how the decision was made. Explainability provides clarity, and that clarity strengthens a person's sense of agency.

Explainability also benefits organizations by identifying weaknesses, revealing unexpected behavior and reducing ethical and operational risks. The article's discussion of intrinsic and post-hoc explainability methods offers structured options for choosing appropriate transparency tools. For organizations committed to accountability, explainability is a practical way to demonstrate responsible decision making.

### **2. What is a "fair" system? Provide an example and describe what achieving fairness would look like.**

A fair system avoids advantaging or disadvantaging people based on irrelevant or sensitive characteristics. The article explains that fairness in Responsible AI requires that outcomes not discriminate based on protected attributes or historical inequities (Barredo Arrieta et al., 2020). Fairness can also be assessed through criteria such as independence or sufficiency, which test how predictions relate to sensitive variables.

It is important to distinguish equality from equity. Equality treats everyone the same, while equity recognizes that people begin from different starting points. Many AI systems produce unequal outcomes when they apply identical rules to individuals with unequal backgrounds. A fair system must consider context, not assume that equal treatment leads to equitable results.

For example, consider an AI system used to assist with graduate program admissions. Applicants do not all have equal access to the experiences that traditionally strengthen applications. Some students grow up with consistent access to food, stable housing, extracurricular opportunities, tutoring, quality schools and extensive support networks. Others face structural barriers that limit these opportunities through no fault of their own. A fair

admissions model would involve evaluating achievement relative to context. This could include weighting indicators of persistence, resourcefulness or success in low-resource environments.

Achieving fairness would involve checking for hidden correlations between opportunity and predicted success, using XAI tools to analyze feature influence and ensuring that the system does not automatically reward privilege. This approach reflects my belief that AI should help surface potential and resilience rather than reinforce preexisting advantages.

### **3. How are bias, fairness, accountability and transparency related?**

Bias refers to distortions in a model's predictions. Fairness is the goal of reducing harmful or irrelevant bias. Transparency and explainability make bias visible so developers, decision makers and affected users can understand how the system behaves (Barredo Arrieta et al., 2020). Accountability ensures that when problems arise, someone is responsible for addressing them.

For me, these concepts also connect to the importance of including diverse perspectives in system design. Diversity and inclusion are not only moral priorities. They improve problem solving and model quality by bringing more perspectives into system evaluation. When the teams designing AI systems reflect the populations they serve, fairness becomes more achievable. Together, bias, fairness, accountability and transparency form the core of Responsible AI because they ensure that systems are understandable, correctable and ethically aligned.

### **4. Give an example of a system that is biased but fair.**

A system can be biased but fair when its bias is directly related to the legitimate purpose of the task. For example, an AI model used by a campus safety office to prioritize building inspections after an earthquake may be biased toward older buildings or structures with known vulnerabilities. This is an intentional and relevant bias because older buildings are at higher risk. The bias does not target demographic groups and instead supports community safety. The article notes that bias is not always harmful and can be necessary when aligned with task-specific requirements (Barredo Arrieta et al., 2020).

Another example is a medical triage system used in emergency care. These systems often prioritize patients with the most severe conditions first, even if other patients arrived earlier. The model is biased toward severity, but this bias is fair because it reflects ethical obligations in healthcare. Its purpose is to save the greatest number of lives, not to preserve a strict first come, first served order or favor those who can pay the most. In this case, the bias directly supports the intended outcome of maximizing patient well-being.

## References

- Barredo Arrieta, A., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., Garcia, S., Gil-Lopez, S., Molina, D., Benjamins, R., Chatila, R., & Herrera, F. (2020). Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 58, 82–115.  
<https://doi.org/10.1016/j.inffus.2019.12.012>